The Pais annular cavity

US 10,144,532 B2 as a Kron ring line, and the patent's figures against physical limits — register entry H18

kronspice simulation report · 2026-11-07 · 1 MW drive scaling, copper walls

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Summary

S. C. Pais's patent "Craft using an inertial mass reduction device" (U.S. Navy, filed 2016, issued 2018) claims that an annular resonant cavity between two hull walls, filled with xenon, driven by microwave emitters, with the outer wall charged, vibrating and rotated, creates "a local polarized vacuum" and reduces the craft's inertial mass. This example solves that cavity as a closed transmission-line ring with Kron's diakoptics (Happ's two-level method), checks it against MNA and the energy ledger, and sets the patent's figures against physical ceilings. The claim is a hypothesis (H18); the standard prediction is computed beside it.

- **Kron and MNA agree:** 3.3×10^{-10} off resonance; on the $Q \approx 1.5\times10^{4}$ resonance, 5.8×10^{-8} with backward errors 8.4×10^{-14} (Kron) and 2.4×10^{-14} (MNA). The interconnection system has order 16 against 1,280 MNA unknowns.
- **Ledger closes:** input power equals wall loss to 2.5×10^{-10}.
- **1 MW into four in-phase emitters** stores 6.167 J at Q 1.477×10^{4}, with 148.9 kV and 2.978 MV/m at most.
- **Sectional steering does not work on a high-Q ring:** energy fed into one section fills the ring evenly (0.250 per quadrant) on or off resonance; only a lossy ring (Q 7.422) keeps 0.817 of it near the emitter.
- **The patent's figures exceed physical limits:** σ = 1 C/m² needs 1.13×10^{11} V/m (wall pressure 5.65×10^{10} Pa against copper's 2.1×10^{8} Pa); 10^{33} W/m² needs a wall speed of 3×10^{13} c; 30,000 rpm at 2 m bursts a carbon-fibre hoop (margin 3.34).
- **Standard prediction:** no net force on a closed cavity, mass unchanged; beaming all 1 MW out gives at most P/c = 3.34 mN. **Status: untested.**

The claim and the standard prediction

From the patent (as summarized in doc/pais-us10144532-comparison.md):

- An inner and an outer resonant-cavity wall form an annular cavity. Microwave emitters drive it, which makes the outer wall vibrate "in an accelerated mode" and "create a local polarized vacuum" outside it.
- The cavity is filled with a noble gas (xenon); the outer wall is electrically charged; the cavity is rotated "in an accelerated mode".
- Figures: emitters at 300 MHz–300 GHz; cavity Q of 10^{4}–10^{9}; wall vibration at 10^{9}–10^{18} Hz; σ = 50,000 C/m² on a 2 m disc at 30,000 rpm; energy flux $S = f_G\cdot(\sigma^2/\varepsilon_0)\cdot[R_r\cdot\omega + R_v\cdot v + v\cdot R]$, about 10^{24} W/m² for rotation and 10^{33} W/m² at σ = 1 C/m² and 10^{11} Hz; steering by driving sections of the shroud separately.
- The patent gives no wall gap, height, voltage or drive power.

Standard prediction: a closed cavity's fields exert no net time-averaged force on the craft (momentum conservation), the mass is unchanged, and the only way to push is to emit momentum, at most P/c. The gyroscope weight reduction the patent cites (Hayasaka & Takeuchi 1989) was not reproduced by three groups (Faller et al. 1990; Quinn & Picard 1990; Nitschke & Wilmarth 1990).

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The construction

The annulus is a closed parallel-plate TEM line around the craft (Paul §4.2): $L' = \mu_0 d/h$, $C' = \varepsilon_0 h/d$, $R' = 2R_s/h$, with d the gap, h the wall height and R_s the copper surface resistance at the drive frequency. Because the patent gives no dimensions, the geometry is declared. Emitters are current injections. It is the same circuit as examples/ring_towers and examples/double_hull: the closed ring transmission line at craft scale and microwave frequency.

parameter	value
radius R	2 m (the patent's example radius)
gap d, wall height h	5 cm, 0.5 m
sections	640 (40 per wavelength)
ring mode	n = 16 at 381.315 MHz (λ = 0.786 m), inside the patent's 300 MHz–300 GHz band
characteristic impedance Z_0	37.7 Ω
walls	copper (case E: niobium, R_s = 20 nΩ, declared)
total drive	1 MW

Kron analysis. Happ's two-level diakoptics with a single common bus (kron.TwoLevel; Happ 1971 Ch. XII §4, Fig. 12.12): the ring is torn into 16 sectors, each hinged at the inner wall, joined by 16 link branches, so the interconnection system has order 16 against 1,280 MNA unknowns. The twin is the same netlist through kronspice's MNA AC analysis.

Results

Five drive cases (1 MW scaling; "quadrant" is the energy fraction in each 90° quarter of the ring, the emitter's quadrant first):

case	f [MHz]	Q	power in	stored energy
A. 4 emitters at 0/90/180/270°, in phase	381.3153	1.477×10^{4}	1×10^{6} W	6.167 J
B. top emitter only, same current, on resonance	381.3153	1.477×10^{4}	6.25×10^{4} W	0.3854 J
C. top emitter only, between modes n and n + 1 (steering attempt)	393.2046	1.5×10^{4}	0.1845 W	1.12×10^{-6} J
D. top emitter only, walls 2000× copper's R_s	381.3153	7.422	107 W	3.315×10^{-7} J

case	V_{max}	E_{max}	RF σ	energy by quadrant	ledger residual
A	148.9 kV	2.978 MV/m	26.4 µC/m²	0.250 0.250 0.250 0.250	2.5×10^{-10}
B	37.22 kV	0.7445 MV/m	6.59 µC/m²	0.250 0.250 0.250 0.250	2.7×10^{-10}
C	0.06346 kV	0.001269 MV/m	0.0112 µC/m²	0.250 0.249 0.250 0.251	1.6×10^{-14}
D	0.06388 kV	0.001278 MV/m	0.0113 µC/m²	0.817 0.089 0.006 0.088	1.0×10^{-14}

Case E, niobium walls with R_s = 20 nΩ (declared; superconducting RF cavities at 2 K): Q 3.76×10^{9}, and the same 6.17 J stored needs 3.93 W.

- **Sectional steering does not work on a high-Q ring.** One wave decays over $1/\alpha = 2Z_0/R' \approx 3.7$ km and the ring is 12.6 m around, so energy fed into one section fills the ring evenly on or off resonance (cases B and C). Only a ring with single-digit Q keeps the energy near its emitter (case D), as the ring-tower runs found: synchronized drives move the antinodes, not which sector holds the energy.
- Driving between modes (case C) does not steer the energy either; it only lowers the power the ring accepts, to 0.1845 W at the same current.

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The patent's figures against physical ceilings (copper case, 1 MW scaling)

figure	value	limit
$\sigma = 1$ C/m²	$E = \sigma/\varepsilon_0 = 1.13\times10^{11}$ V/m	Kilpatrick 1.91×10^{7} V/m at 381 MHz; Fowler–Nordheim emission 2.45×10^{15} A/m² drains the charge in 4.1×10^{-16} s; wall pressure 5.65×10^{10} Pa against copper's 2.1×10^{8} Pa (UTS); RF drive to reach it in this cavity 1.44×10^{15} W
$\sigma = 50{,}000$ C/m²	$E = 5.65\times10^{15}$ V/m	4×10^{-3} of the Schwinger field 1.32×10^{18} V/m; pressure 1.41×10^{20} Pa; emission 1.09×10^{25} A/m² drains it in 4.6×10^{-21} s; drive 3.6×10^{24} W
10^{24} W/m² (rotation)	reproduced: $\sigma^2/\varepsilon_0\cdot R\omega = 1.77\times10^{24}$ W/m²	the magnitude of E×H beside a moving charged surface; it circulates (div S = 0), so no energy leaves
10^{33} W/m² (vibration)	needs $R_v\cdot v = 8.85\times10^{21}$ m/s	3×10^{13} c
30,000 rpm at R = 2 m	rim 6,283 m/s, $(v/c)^2 = 4.4\times10^{-10}$	burst margin 3.34 against a T1000G carbon-fibre hoop (spacetime.AnalyzeRotor; > 1: flies apart)
wall vibration 10^{9}–10^{18} Hz	—	lattice vibrations end at the Debye frequency: Cu 7.1×10^{12} Hz, Al 8.9×10^{12} Hz
thrust	—	zero net time-averaged force on a closed cavity; beaming all 1 MW out gives P/c = 3.34 mN

Twin tests and ledger

check	result
Kron vs MNA at 393.2046 MHz (off resonance)	3.3×10^{-10} relative; backward error Kron 3.9×10^{-16}, MNA 1.3×10^{-13}
Kron vs MNA at 381.3153 MHz (on resonance)	5.8×10^{-8} relative; backward error Kron 8.4×10^{-14}, MNA 2.4×10^{-14}
ledger (input − wall loss)/input	1.0×10^{-14} to 2.7×10^{-10} over cases A–D
Q against ωL′/R′	to 2×10^{-3}
resonance against the lumped-ladder dispersion	follows sin x/x exactly
export sheet, case A (torn into 16 pieces of 119–120 unknowns)	largest node-voltage difference 1.1×10^{-8} of the largest node voltage; power taken by the R, L, C branches equal torn and untorn (Roth 1959 §5)

On the resonance the system is ill-conditioned by about Q·n, so both solutions are checked by their backward error rather than by their difference alone.

Limits of the model

- The geometry (gap, height, drive power) is declared; the patent gives none of it.
- The cavity is a 1-D TEM ring line with copper surface loss; the xenon fill, the static charge, the rotation and the wall vibration are assessed against limits, not simulated.
- Emitters are ideal current injections.

Status and bench test

H18: untested. Bench test: weigh a sealed, microwave-driven, charged cavity on a balance against a dummy load dissipating the same power (to separate heating and convection). The standard prediction is no weight change beyond P/c of any escaping radiation, with every watt booked in through the emitters and out as heat.

Reproducing the results

```
go run ./examples/pais_cavity
go test ./examples/pais_cavity
KRONSPICE_EXPORT=1 go test ./examples/pais_cavity -run TestExport   # export/ sheet for case A
```


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